

Figure 1: Riesz iteration: $M_n = \mathrm{span}\{x_1, \ldots, x_n\}$, $\mathrm{dist}(x_{n+1}, M_n) > \frac{1}{2} \Rightarrow \|x_i - x_j\| > \frac{1}{2}$ $(i \neq j)$; hence no Cauchy subsequence in $\overline{B}_{\mathcal{X}}$.